MCRS 4-TOWER PLATFORM: 90-DOCUMENT IP TREE

Master Intellectual Property Package & Engineering Transfer Index (Electric & Hybrid)

DOCUMENT REF	IP SCOPE	TOTAL DELIVERABLES	TRANSFER LEVEL
MCRS-IP-TREE-90-v1.0	**4T-Electric & 4T-Hybrid**	**90 Core Technical Docs**	**Full Commercial / OEM**

IP Transfer Framework Summary: This master index specifies the complete 90-document technical baseline required to fully license, manufacture, certify, and transfer the MCRS 4-Tower System (50-Ton Electric & 80-Ton Hybrid). Each document contains explicit dual-side tracking numbers (#1 to #90) ensuring complete traceability across technical, mathematical, control, thermal, and legal domains.

TIER 1: SYSTEM BASELINE & MASTER ARCHITECTURE (1–10)

System Requirements & Specifications

#	Document ID	Document Title	Document Type	Gate	#
1	MCRS-SYS-SPEC-001	**MCRS 4-Tower Platform Master System Specification**	System Specification	G0/G1	1
2	MCRS-SYS-ICD-002	**Master Interface Control Document (ICD) - Host & Rigging**	Interface Specification	G1	2
3	MCRS-SYS-CONOPS-003	**Concept of Operations (CONOPS) - Heavy Industrial Rigging**	Operational Specification	G0	3
4	MCRS-SYS-REQ-004	**System Safety Requirements & Hazard Classification (ISO 13849)**	Safety Requirement	G1	4
5	MCRS-SYS-ENV-005	**Environmental & Harsh Climate Operating Baseline (-40°C to +55°C)**	Environmental Spec	G1	5
6	MCRS-SYS-ARCH-006	**4-Tower Modular Kinematic Architecture & Load Path Topology**	Architecture Doc	G2	6
7	MCRS-SYS-MASS-007	**System Mass Properties, CG Tracking & Weight Budget Master**	Technical Report	G2	7
8	MCRS-SYS-RAMS-008	**RAMS Analysis Report (Reliability, Availability, Maintainability)**	Analytical Report	G3	8
9	MCRS-SYS-STD-009	**Regulatory Compliance Matrix (EN 280, ASME B30.1, CE, UL)**	Compliance Matrix	G4	9
10	MCRS-SYS-VER-010	**Master Verification & Validation Plan (V&V Master)**	Test Master Plan	G5	10

TIER 2: MASTER MECHANICAL & TOWER STRUCTURE (11–20)

Mechanical & CAD Design Baseline

#	Document ID	Document Title	Document Type	Gate	#
11	MCRS-MEC-DWG-011	**Master Tower Assembly Drawings & GD&T Package (2D/3D)**	CAD / Drawing Package	G2	11
12	MCRS-MEC-PRS-012	**Planetary Roller Screw Selection, Load Rating & Sizing Specification**	Component Spec	G2	12
13	MCRS-MEC-GUD-013	**Telescopic Guide Rail & Bronze-Graphite Wear Plate Specification**	Component Spec	G2	13
14	MCRS-MEC-OUT-014	**Hydraulic Outrigger Assembly & Ground Pad Structural Drawings**	Drawing Package	G3	14
15	MCRS-MEC-LCK-015	**Fail-Safe Mechanical Locking Pin Mechanism Specification**	Mechanical Design	G3	15

TIER 2: MASTER MECHANICAL & TOWER STRUCTURE (CONTINUED: 16–20)

Mechanical & Manufacturing

#	Document ID	Document Title	Document Type	Gate	#
16	MCRS-MEC-MAT-016	Structural Steel Alloy Specification (S960QL / Weldox 900)	Material Specification	G2	16
17	MCRS-MEC-WLD-017	Welding Procedure Specification (WPS) & Quality Plan (ISO 3834)	Manufacturing Spec	G3	17
18	MCRS-MEC-GBX-018	High-Torque Planetary Reduction Gearbox Sizing & Lubrication Plan	Component Spec	G3	18
19	MCRS-MEC-RIG-019	Quick-Release Load Beam Interface & Equalization Rigging Spec	Rigging Design	G4	19
20	MCRS-MEC-BOM-020	Master Mechanical Bill of Materials (BOM) with Sourcing Codes	Manufacturing BOM	G4	20

TIER 3: 4T-ELECTRIC SYSTEM ARCHITECTURE (50-TON BEV) (21–30)

800V BEV High-Voltage Package

#	Document ID	Document Title	Document Type	Gate	#
21	MCRS-ELE-ARCH-021	800V DC Architecture Specification & Electrical Single-Line Diagram	Electrical Spec	G1	21
22	MCRS-ELE-MOT-022	Permanent Magnet Synchronous Motor (PMSM) Servo Sizing & Spec	Component Spec	G2	22
23	MCRS-ELE-INV-023	SiC High-Voltage Dual Inverter Drive Specification & Control ICD	Drive Component Spec	G2	23
24	MCRS-ELE-BAT-024	Lithium LFP Battery Energy Storage System (BESS) Integration Report	BESS Integration Spec	G2	24
25	MCRS-ELE-BMS-025	Battery Management System (BMS) Safety, Balancing & ISO 26262 Plan	BMS Safety Document	G3	25
26	MCRS-ELE-REG-026	Regenerative Braking & Descent Energy Recovery Strategy	Energy Analysis	G3	26
27	MCRS-ELE-PDU-027	High Voltage Power Distribution Unit (HV-PDU) Schematics	Schematic Package	G3	27
28	MCRS-ELE-HAR-028	HV Harness Cable Routing, Shielding & Connector Specification	Harness Design Spec	G4	28
29	MCRS-ELE-EMC-029	EMC/EMI Shielding Analysis & Compliance Protocol (IEC 61000)	Compliance Test Plan	G4	29
30	MCRS-ELE-BOM-030	Master Electrical Bill of Materials (4T-Electric)	Manufacturing BOM	G4	30

TIER 4: 4T-HYBRID SYSTEM ARCHITECTURE (80-TON ELECTRO-HYDRAULIC) (31–35)

Electro-Hydraulic & Engine Package

#	Document ID	Document Title	Document Type	Gate	#
31	MCRS-HYB-SCH-031	Master Electro-Hydraulic Circuit Schematic & Valve Topology	Hydraulic Schematic	G1	31
32	MCRS-HYB-PMP-032	Variable Displacement Axia-Piston Pump & Power Match Report	Hydraulic Design	G2	32
33	MCRS-HYB-ENG-033	Tier 4F Diesel Engine Integration & Power Take-Off (PTO) Spec	Powerplant Integration	G2	33
34	MCRS-HYB-ACC-034	Piston Accumulator Sizing & Hydraulic Energy Recovery Strategy	Hydraulic Design	G2	34
35	MCRS-HYB-VAL-035	Proportional Valve Block Design & Counterbalance	Component Spec	G3	35

TIER 4: 4T-HYBRID SYSTEM ARCHITECTURE (CONTINUED: 36–40)

Fluid & Hybrid Power

#	Document ID	Document Title	Document Type	Gate	#
36	MCRS-HYB-FLD-036	**Synthetic Biodegradable Hydraulic Fluid Baseline & Cold-Start Spec**	Fluid Specification	G3	36
37	MCRS-HYB-MAN-037	**Custom Hydraulic Manifold 3D CAD Block & Machining Drawings**	CAD / Manufacturing	G3	37
38	MCRS-HYB-FIL-038	**Hydraulic Filtration & Contamination Monitoring Specification**	Maintenance Spec	G4	38
39	MCRS-HYB-PIP-039	**High-Pressure Piping, Hose Assembly & Swivel Joint Layout**	Routing Spec	G4	39
40	MCRS-HYB-BOM-040	**Master Hydraulic Bill of Materials (4T-Hybrid)**	Manufacturing BOM	G4	40

TIER 5: CONTROL SYSTEMS, SOFTWARE & SAFETY (41–50)

Embedded Software & Controls

#	Document ID	Document Title	Document Type	Gate	#
41	MCRS-CTL-ARC-041	**Triple-Redundant Safety PLC Architecture & CANopen Topology**	Controls Architecture	G1	41
42	MCRS-CTL-SYN-042	**Closed-Loop 4-Tower Synchronization Control Algorithm Spec**	Control Algorithm Spec	G2	42
43	MCRS-CTL-SWS-043	**Software Requirement Specification (SRS) - Functional Safety SIL 3**	Software Specification	G2	43
44	MCRS-CTL-HMI-044	**Operator Wireless Remote & Graphical HMI Interface ICD**	HMI Interface Spec	G2	44
45	MCRS-CTL-SEN-045	**Multi-Axis Position, Load Cell & Tilt Sensor Integration Spec**	Sensor Specification	G3	45
46	MCRS-CTL-FLT-046	**Fault Detection, Isolation & Emergency Shutdown Logic Tree**	Safety Logic Spec	G3	46
47	MCRS-CTL-CYB-047	**Cybersecurity Assessment & Encrypted Wireless Protocol (IEC 62443)**	Cybersecurity Spec	G3	47
48	MCRS-CTL-HIL-048	**Hardware-in-the-Loop (HIL) Software Test Report & Validation**	Test Report	G4	48
49	MCRS-CTL-SRC-049	**Firmware & Control Application Source Code Repository Index**	Software IP Package	G4	49
50	MCRS-CTL-CAL-050	**Control Loop Calibration & Parameter Field Tuning Manual**	Field Operations Spec	G5	50

TIER 6: STRUCTURAL, FEA & KINEMATIC ENGINEERING (51–55)

Structural & Dynamics Physics

#	Document ID	Document Title	Document Type	Gate	#
51	MCRS-FEA-STA-051	**Static Finite Element Analysis (FEA) - 100% Rated Payload**	FEA Engineering Report	G2	51
52	MCRS-FEA-DYN-052	**Dynamic FEA & Dynamic Magnification Factor (SF_d) Analysis**	FEA Dynamic Report	G2	52
53	MCRS-FEA-BCK-053	**Multi-Stage Non-Linear Euler & Johnson Buckling Analysis**	Mathematical Report	G2	53
54	MCRS-FEA-FAT-054	**Fatigue & Weld Joint Life Prediction (L_{10h}) Analysis**	Fatigue Life Study	G3	54
55	MCRS-FEA-WND-055	**Outdoors Wind Load & Seismic Stability Calculation (ASCE 7)**	Structural Report	G3	55

TIER 6: STRUCTURAL & FEA ENGINEERING (CONTINUED: 56–60)

Structural Simulation

#	Document ID	Document Title	Document Type	Gate	#
56	MCRS-FEA-ASY-056	Asymmetric Payload Loading & Torsional Misalignment Study	Simulation Report	G3	56
57	MCRS-FEA-GRD-057	Ground Reaction Force & Outrigger Pad Soil Bearing Study	Geotechnical Report	G3	57
58	MCRS-FEA-IMP-058	Side Impact, Collision & Emergency Drop Shock Analysis	Dynamic Simulation	G4	58
59	MCRS-FEA-BOL-059	High-Strength Structural Bolted Joint Verification (VDI 2230)	Analytical Calculation	G4	59
60	MCRS-FEA-COR-060	FEA Correlation & Strain Gauge Test Validation Report	V&V Report	G5	60

TIER 7: THERMAL, FLUID & ELECTRICAL PHYSICS MODELS (61–70)

CFD & Multi-Physics

#	Document ID	Document Title	Document Type	Gate	#
61	MCRS-THM-BAT-061	BESS Thermal Management CFD Model & Liquid Cooling Sizing	CFD Analysis	G2	61
62	MCRS-THM-INV-062	Power Electronics Inverter Junction Temperature Analysis	Thermal Report	G2	62
63	MCRS-THM-HYD-063	Hydraulic Oil Heat Exchanger Sizing (-40°C to +55°C Range)	Thermal Calculation	G2	63
64	MCRS-THM-MOT-064	Servo Motor Continuous vs Peak Duty Thermal Characterization	Test & Math Report	G3	64
65	MCRS-THM-COL-065	Extreme Cold Extreme Weather Pre-Heating System Specification	System Specification	G3	65
66	MCRS-THM-CFD-066	Cabinet Enclosure Internal Airflow & Dust Filtration CFD	CFD Report	G3	66
67	MCRS-THM-LUB-067	Planetary Screw Thermal Expansion & Lubricant Degradation	Tribology Report	G4	67
68	MCRS-THM-LNS-068	Loss Network Model & System Efficiency Map (Electric vs Hybrid)	Math Model Report	G4	68
69	MCRS-THM-ARC-069	High Voltage Arc Fault Thermal Hazard Analysis (NFPA 70E)	Safety Analysis	G4	69
70	MCRS-THM-VLD-070	Thermal Chamber Environmental Test Validation Report	Test Report	G5	70

TIER 8: VERIFICATION PROTOCOLS & PROOF BENCH MARKS (71–75)

Factory & Field Testing

#	Document ID	Document Title	Document Type	Gate	#
71	MCRS-TST-FAT-071	Factory Acceptance Test (FAT) Protocol & Acceptance Criteria	Testing Protocol	G3	71
72	MCRS-TST-SAT-072	Site Acceptance Test (SAT) & Client Commissioning Procedure	Testing Protocol	G4	72
73	MCRS-TST-PRF-073	125% Static & 110% Dynamic Proof Load Test Standard Procedure	Proof Test Spec	G4	73
74	MCRS-TST-END-074	500-Cycle Endurance & Accelerated Life Testing (ALT) Protocol	Endurance Test Plan	G4	74
75	MCRS-TST-EMG-075	Emergency Power Cut & Hose Rupture Drop Test Protocol	Safety Test Spec	G4	75

TIER 8: VERIFICATION PROTOCOLS & TESTING (CONTINUED: 76–80) Certification Protocols

#	Document ID	Document Title	Document Type	Gate	#
76	MCRS-TST-SYN-076	**4-Tower Synchronization Accuracy & ±1.0mm Drift Measurement**	Test Procedure	G4	76
77	MCRS-TST-NVH-077	**Noise, Vibration & Harshness (NVH) Field Measurement Report**	NVH Report	G5	77
78	MCRS-TST-CE-078	**CE Technical Construction File (TCF) & Machinery Directive Dossier**	Certification File	G5	78
79	MCRS-TST-UL-079	**UL 2202 / UL 2580 High Voltage Electrical Safety Dossier**	Certification File	G5	79
80	MCRS-TST-REP-080	**Independent Third-Party Inspection & Type Approval Certificate**	Certificate of Approval	G5	80

TIER 9: IP LEGAL, LICENSING & PRODUCTION PACKAGE (81–90) Legal & Technology Transfer

#	Document ID	Document Title	Document Type	Gate	#
81	MCRS-IP-PAT-081	**Master Patent Claims & Freedom to Operate (FTO) Legal Matrix**	Legal IP File	G0	81
82	MCRS-IP-LIC-082	**Technology Transfer Agreement & IP Licensing Terms Template**	Legal Contract	G1	82
83	MCRS-IP-TKN-083	**Know-How & Trade Secret Protection Register (Process IP)**	IP Register	G2	83
84	MCRS-PRD-SUP-084	**Approved Vendor List (AVL) & Tier-1 Global Supply Chain Directory**	Supply Chain Spec	G3	84
85	MCRS-PRD-JIG-085	**Manufacturing Tooling, Assembly Jigs & Fixture CAD Package**	Manufacturing CAD	G3	85
86	MCRS-PRD-SOP-086	**Standard Operating Procedures (SOP) for Factory Assembly**	Manufacturing SOP	G4	86
87	MCRS-PRD-OAM-087	**Master Operation & Maintenance Manual (4T-Electric & Hybrid)**	User Manual	G4	87
88	MCRS-PRD-PAR-088	**Illustrated Spare Parts Catalog & Recommended Field Service Stock**	Parts Catalog	G5	88
89	MCRS-PRD-CST-089	**Bill of Materials Costing Model & Target Production COGS Analysis**	Financial Model	G5	89
90	MCRS-IP-SIG-090	**Master Technology Transfer Handover & Quality Sign-Off Record**	Transfer Handover	G5	90

IP Transfer Sign-Off & Delivery Status: All 90 documents listed in this master index constitute the total intellectual property and technical transfer package for the MCRS 4-Tower Platform. Full compliance with international engineering standards (ISO 13849, ASME B30.1, EN 280, UL 2580) is built directly into every document tier.